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Vadodara, Gujarat
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CSE

Unit-1: Introduction to Natural Language Processing

Mr.HIMANSHU TIWARI
Deep Learning with NLP(303105479)
Artificial Intelligence and Data Science Department

Content

- NLP- Introduction
- First Wave, Second Wave and Third Wave
- NLP- Difficulty, Importance
- Basic NLP Operations
    - Word Level Analysis
    - Syntactic Analysis
    - Semantic Analysis
    - Word Sense Disambiguation
    - Discourse Processing,
    - Part of Speech (PoS) Tagging

• Natural Language Inception, Information Retrieval

•Applications of NLP

Introduction to Natural Language Processing (NLP)

Introduction:

- Natural Language Processing (NLP) is a field of artificial intelligence (AI) that focuses on the interaction between computers and human language.
- NLP has evolved through different waves, each emphasizing different approaches and techniques.
- In this presentation, we will explore the three waves of NLP: Rationalism, Empiricism, and Deep Learning.

Introduction to Natural Language Processing (NLP)

First Wave: Rationalism

- The Rationalism wave occurred in the early years of NLP, from the 1950s to the 1980s.
- Key idea: Understanding language through formal rules and logical reasoning.
- Symbolic approaches: Developed knowledge-based systems using handcrafted rules and linguistic frameworks.
- Major challenge: Capturing the complexity and ambiguity of natural language.

Introduction to Natural Language Processing (NLP)

First Wave Techniques:

- Rule-based systems: Explicitly defined linguistic rules and grammatical structures.
- Expert systems: Knowledge-based systems using domain-specific rules and inference engines.
- Semantic analysis: Focus on representing the meaning of words and sentences using logical forms.
- Limitations: Difficulty in handling ambiguity, lack of scalability, and the need for extensive manual knowledge engineering.

Introduction to Natural Language Processing (NLP)

Second Wave: Empiricism

- The Empiricism wave emerged in the 1990s and aimed to address the limitations of the Rationalism wave.
- Key idea: Language understanding through statistical and machine learning approaches.
- Corpus-based methods: Leveraged large-scale text data for training and modeling language patterns.
- Statistical techniques: Used probabilistic models and algorithms for language processing tasks.

Introduction to Natural Language Processing (NLP)

Second Wave Techniques:

- Part-of-speech tagging: Assigning grammatical categories to words.
- Named Entity Recognition (NER): Identifying and classifying named entities in text.
- Statistical machine translation: Translating text from one language to another using statistical models.
- Sentiment analysis: Determining the sentiment or emotion expressed in text.
- Challenges: Reliance on annotated data, sparse feature representations, and limitations in capturing semantic meaning.

Introduction to Natural Language Processing (NLP)

Third Wave: Deep Learning

- The Deep Learning wave emerged in the 2010s, driven by advancements in computational power and data availability.
- Key idea: Leveraging neural networks with multiple layers to learn hierarchical representations of language.
- Neural network architectures: Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and Transformers.
- Deep learning models: Sequence-to-sequence models, attention mechanisms, and language models.

Introduction to Natural Language Processing (NLP)

Third Wave Techniques:

•Word embeddings: Transforming words into dense vector representations to capture semantic relationships.

•Language generation: Generating coherent and contextually relevant text using models like GPT-3.

•Question answering: Providing precise answers to questions based on comprehension of text.

•Sentiment analysis: Predicting sentiment or emotion in text with improved accuracy.

•Challenges: Data requirements, model interpretability, and ethical considerations.

What Makes Natural Language Processing Difficult?

Natural Language Processing (NLP) presents several challenges that make it a difficult task. Here are some factors contributing to the complexity of NLP:

1. Ambiguity: Natural language is inherently ambiguous, and words or sentences can have multiple interpretations depending on the context. Resolving this ambiguity requires understanding the meaning and intent behind the language.

2. Context Dependency: The meaning of words and phrases can change based on the surrounding context. NLP models need to consider the broader context to accurately interpret and process language.

3. Syntax and Grammar Variations: Natural languages exhibit a wide range of syntactic and grammatical variations across different dialects, languages, and writing styles. NLP systems must account for these variations to ensure robust language understanding.

What Makes Natural Language Processing Difficult?

4. Named Entity Recognition: Identifying and classifying named entities (such as names of people, organizations, locations) in text is challenging due to the vast number of possible variations and the lack of consistent naming conventions.

5. Language Understanding: Extracting the intended meaning from text requires understanding nuances, idioms, sarcasm, humor, and other forms of implicit language.

6. Lack of Standardization: Language use is constantly evolving, and there is a lack of standardized rules or conventions for many aspects of language. This poses difficulties in creating comprehensive language models that can handle all linguistic variations.

7. Data Sparsity: Obtaining labeled training data for NLP tasks can be challenging, especially for specialized domains or low-resource languages. Limited data availability can impact the performance and generalization of NLP models.

8. Real-world Noise: NLP systems often encounter noisy and incomplete data, including misspellings, grammatical errors, abbreviations, and informal language usage.

Importance of Natural Language Processing (NLP)

Following are the importance area where the NLP is used:

1. Language Understanding
2. Information Extraction
3. Text Summarization
4. Machine Translation
5. Speech Recognition and Synthesis
6. Question Answering
7. Dialogue Systems
8. Knowledge Representation

NLP Basic Terminologies

- **Phonology:** It is study of organizing sound systematically.
- **Morphology:** The study of the formation and internal structure of words.
- **Morpheme:** It is primitive unit of meaning in a language.
- **Syntax:** The study of the formation and internal structure of sentences.
- **Semantics:** The study of the meaning of sentences.
- **Pragmatics:** It deals with using and understanding sentences in different situations and how the interpretation of the sentence is affected.
- **Discourse:** It deals with how the immediately preceding sentence can affect the interpretation of the next sentence.
- **World Knowledge:** It includes the general knowledge about the world.

Important Steps/Basic NLP Operations:

1. Word Level Analysis (Lexical Analysis)
2. Syntactic Analysis (Parsing)
3. Semantic Analysis
4. Discourse Integration
5. Pragmatic Analysis

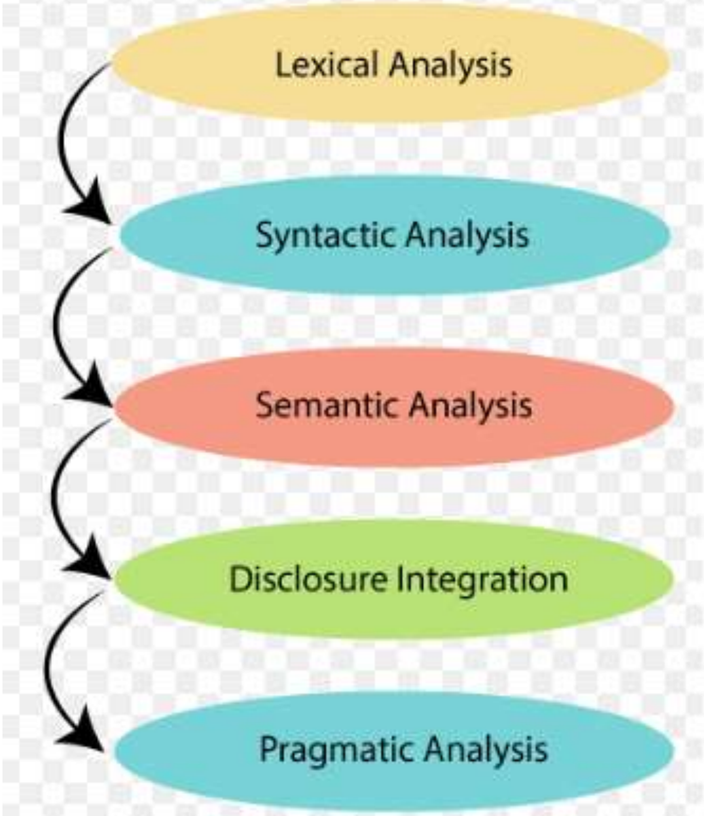


Figure-1: Stages of NLP

Important Steps/Basic NLP Operations:

1. **Word Level Analysis (Lexical Analysis):**
   - This stage focuses on analyzing individual words or tokens in a text.
   - It involves tasks like tokenization (breaking text into smaller units), part-of-speech tagging (assigning grammatical categories to words), lemmatization (reducing words to their base form), and stemming (reducing words to their root form).
   - Word level analysis helps in understanding the basic components and properties of the text.

2. **Syntactic Analysis (Parsing):**
   - Syntactic analysis involves understanding the structure and grammar of a sentence or text.
   - It includes tasks like parsing, where the relationships between words are identified and represented in a syntactic tree or graph.
   - This stage helps in determining the grammatical structure, identifying phrases and clauses, and analyzing the dependencies between words.

Important Steps/Basic NLP Operations:

3. Semantic Analysis:

•Semantic analysis focuses on understanding the meaning and interpretation of a sentence or text.

• It involves tasks like word sense disambiguation (resolving multiple possible meanings of words), semantic role labeling and semantic parsing (representing the meaning of sentences in a structured form).

•Semantic analysis helps in capturing the intended meaning and semantic relationships within the text.

4. Discourse Integration:

•Discourse integration involves analyzing the larger context and coherence of a text beyond individual sentences.

•It focuses on understanding how sentences and ideas connect and relate to each other to form a coherent discourse.

•Tasks in this stage include identifying and linking pronouns to their referents, anaphora resolution, and discourse parsing. Discourse integration ensures the overall coherence and flow of meaning within a text.

Important Steps/Basic NLP Operations:

5. Pragmatic Analysis:

- Pragmatic analysis involves understanding the pragmatic aspects of language use, including the speaker's intentions, implicatures, and the context in which the language is used.
- It aims to capture the intended meaning beyond the literal interpretation of the text. Pragmatic analysis takes into account factors like sarcasm, irony, politeness, and cultural nuances that affect the meaning and interpretation of the text.

These stages in NLP progressively build upon each other, starting from analyzing individual words and their properties, moving on to understanding the structure and grammar of sentences, capturing the meaning and semantics of the text, integrating discourse-level information, and considering the pragmatic aspects of language use.

Part-of-speech (POS) Tagging

- Part-of-speech (POS) tagging is a fundamental task in Natural Language Processing (NLP) that involves assigning grammatical categories or tags to words in a sentence. The tags represent the syntactic role and part of speech of each word, such as noun, verb, adjective, adverb, pronoun, preposition, conjunction, etc.
- POS tagging plays a crucial role in various NLP applications, including language understanding, text analysis, information extraction, and machine translation. Here's an overview of how POS tagging works:

1. Corpus and Tagset: POS tagging models are trained on annotated corpora, which are large collections of text with each word tagged with its respective part of speech. A tagset is a predefined set of tags that the model assigns to words.

Part-of-speech (POS) Tagging

2.Training: To build a POS tagging model, machine learning algorithms, such as Hidden Markov Models (HMM), Maximum Entropy Markov Models (MEMM), or neural network-based models like Recurrent Neural Networks (RNNs) are trained on the annotated corpus. The models learn statistical patterns and contextual information to predict the appropriate tag for each word.

3. Feature Extraction: During training, various features are extracted from the input word and its surrounding context to provide information for the model's prediction. These features may include the word itself, neighboring words, prefixes, suffixes, capitalization, and other contextual information.

4. Tagging Process: Once trained, the POS tagging model can be used to assign tags to unseen text. The input sentence is tokenized into individual words, and the model predicts the most likely POS tag for each word based on the learned patterns and contextual information.

5. Tagset Evaluation: The accuracy of POS tagging is typically evaluated by comparing the predicted tags with the gold standard tags in the annotated corpus. Metrics such as precision, recall, and F1 score are commonly used to assess the performance of the POS tagging model

English Part-of-speech (POS)

- Original Brown corpus used a large set of 87 POS tags.
- Most common in NLP today is the Penn Treebank set of 45 tags.
  - Tagset used in these slides.
  - Reduced from the Brown set for use in the context of a parsed corpus (i.e. treebank).
- The C5 tagset used for the British National Corpus (BNC) has 61 tags.

English Part-of-speech (POS)

Noun (person, place or thing)

- Singular (NN): dog, fork
- Plural (NNS): dogs, forks
- Proper (NNP, NNPS): John, Springfields
- Personal pronoun (PRP): I, you, he, she, it
- Wh-pronoun (WP): who, what

Verb (actions and processes)

- Base, infinitive (VB): eat
- Past tense (VBD): ate
- Gerund (VBG): eating
- Past participle (VBN): eaten
- Non 3rd person singular present tense (VBP): eat
- 3rd person singular present tense: (VBZ): eats
- Modal (MD): should, can
- To (TO): to (to eat)

English Part-of-speech (POS)

- Adjective (modify nouns)
  - Basic (JJ): red, tall
  - Comparative (JJR): redder, taller
  - Superlative (JJS): reddest, tallest
- Adverb (modify verbs)
  - Basic (RB): quickly
  - Comparative (RBR): quicker
  - Superlative (RBS): quickest
- Preposition (IN): on, in, by, to, with
- Determiner:
  - Basic (DT) a, an, the
  - WH-determiner (WDT): which, that
- Coordinating Conjunction (CC): and, but, or,
- Particle (RP): off (took off), up (put up)

English Part-of-speech (POS)

•Annotate each word in a sentence with a part-of-speech marker. Lowest level of syntactic analysis.

•It is useful for subsequent syntactic parsing and word sense disambiguation.

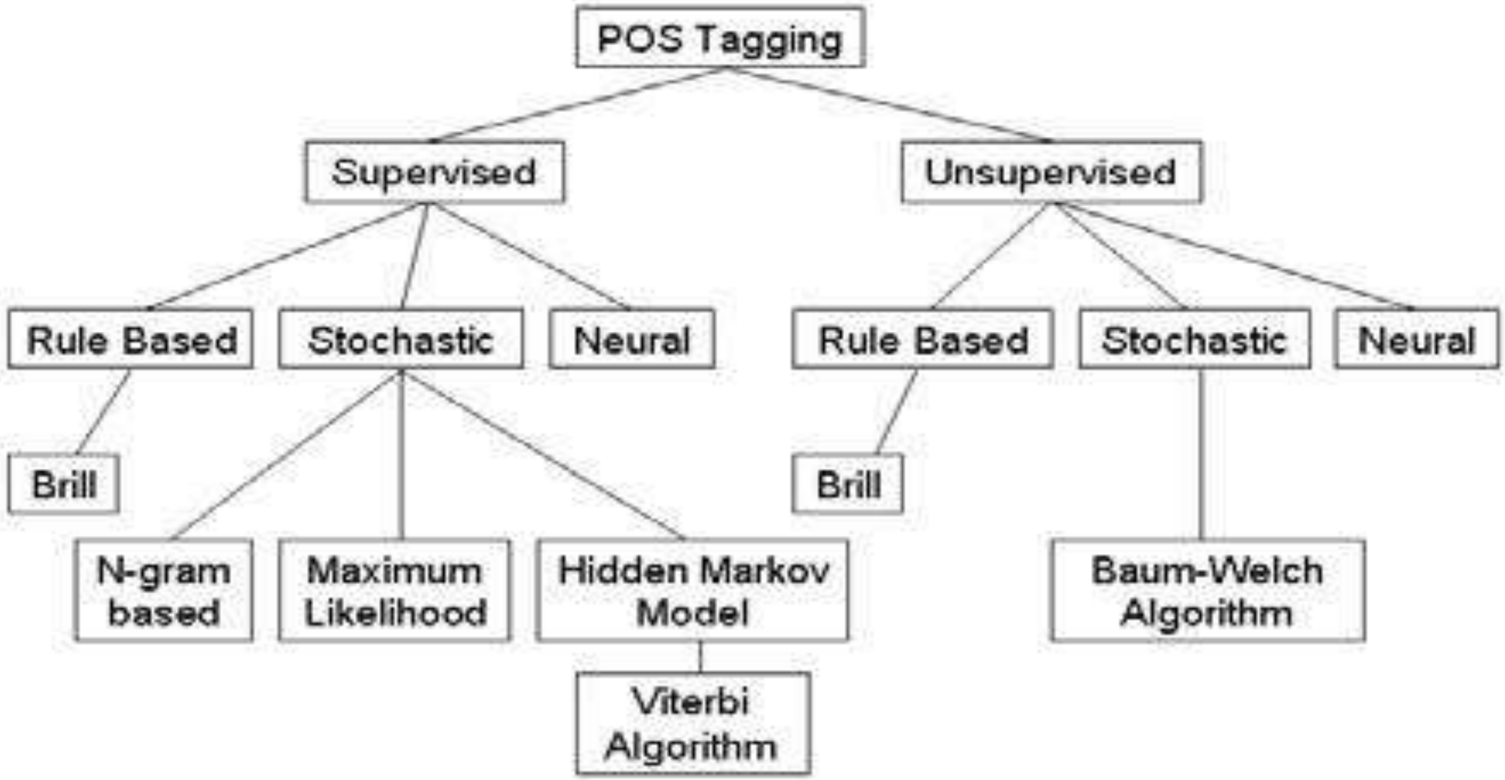
•For Example:

Ram eat mangoes.
NNP VB NNS

POS Tagging Approaches

- **Rule-Based**: Human crafted rules based on lexical and other linguistic knowledge.
- **Learning-Based**: Trained on human annotated corpora like the Penn Treebank.
- **Statistical models**: Hidden Markov Model (HMM), Maximum Entropy Markov Model (MEMM), Conditional Random Field (CRF)
- **Rule learning**: Transformation Based Learning (TBL)
- **Neural networks**: Recurrent networks like Long Short Term Memory (LSTMs)
- Generally, learning-based approaches have been found to be more effective overall, taking into account the total amount of human expertise and effort involved.

POS Tagging Approaches



Classification Learning

Typical machine learning addresses the problem of classifying a feature-vector description into a fixed number of classes.

There are many standard learning methods for this task:

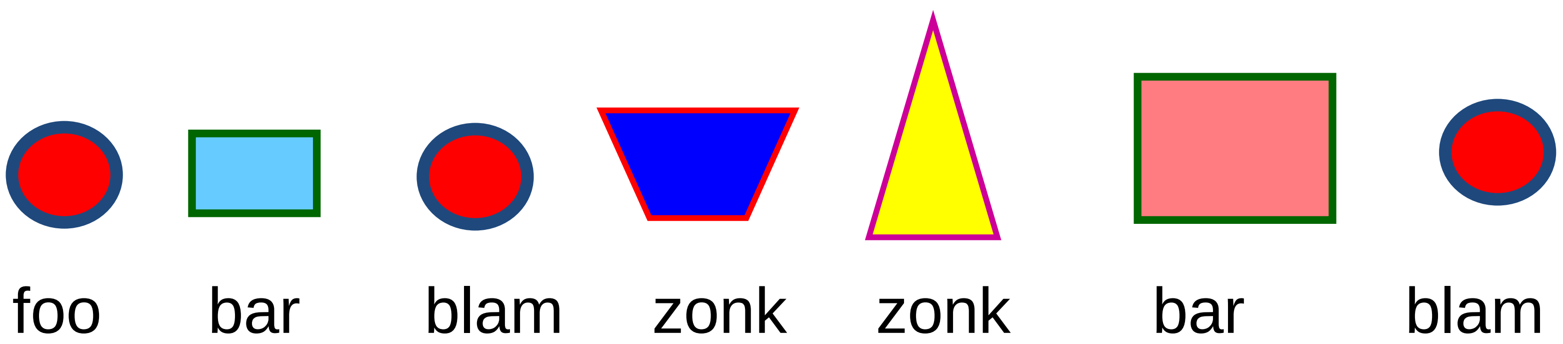
- Decision Trees and Rule Learning
- Naïve Bayes and Bayesian Networks
- Logistic Regression / Maximum Entropy (MaxEnt)
- Perceptron and Neural Networks
- Support Vector Machines (SVMs)
- Nearest-Neighbor / Instance-Based

Beyond Classification Learning

- Standard classification problem assumes individual cases are disconnected and independent (i.i.d.: independently and identically distributed).
- Many NLP problems do not satisfy this assumption and involve making many connected decisions, each resolving a different ambiguity, but which are mutually dependent.
- More sophisticated learning and inference techniques are needed to handle such situations in general.

Sequence Labeling Problem

•Many NLP problems can viewed as sequence labeling.
•Each token in a sequence is assigned a label.
•Labels of tokens are dependent on the labels of other tokens in the sequence, particularly their neighbors (not i.i.d).



Natural Language Inception

- Natural Language Inception refers to the idea of creating a loop or cycle where natural language is used to generate or analyze natural language itself. It involves the use of techniques and models in Natural Language Processing (NLP) to process, understand, and generate human language.
- The concept of Natural Language Inception draws inspiration from the movie "Inception," where dreams are nested within dreams. In the context of NLP, it represents the recursive nature of using language to manipulate and generate language, creating a feedback loop.

There are several ways in which Natural Language Inception can be observed:

1. Language Generation: NLP models can generate human-like text by leveraging techniques such as language modeling or sequence-to-sequence models. This generated text can be further processed and analyzed by other NLP models, creating a loop where language is continually generated and analyzed.

2. Machine Translation: Machine translation involves using NLP models to translate text from one language to another. The translated text can then be fed back into the model to generate translations in other languages, forming a recursive process of language transformation.

3. Question-Answering Systems: Question-answering systems utilize NLP techniques to understand and generate responses to user queries. The answers generated can be in the form of natural language, which can then be further processed or analyzed by other NLP models, allowing for deeper levels of understanding and interaction.

4. Chatbots and Conversational Agents: Chatbots and conversational agents use NLP algorithms to engage in human-like conversations. They analyze user input, generate responses, and adapt their behavior based on the conversation context.

5. Reinforcement Learning in NLP: Reinforcement learning techniques can be applied to NLP tasks, where language models learn to optimize their performance based on feedback.

Information Retrieval (Extraction)

- Identify phrases in language that refer to specific types of entities and relations in text.
- Named entity recognition is task of identifying names of people, places, organizations, etc. in text.
- people organizations places

Michael Dell is the CEO of Dell Computer Corporation and lives in Austin Texas.

- Extract pieces of information relevant to a specific application, e.g. used car ads:

make model year mileage price

For sale, 2002 Toyota Prius, 20,000 mi, $15K or best offer. Available starting July 30, 2006.

Information Retrieval (Extraction)

•Information Retrieval (IR) in Natural Language Processing (NLP) refers to the process of retrieving relevant information from a collection of documents or data based on a user's query. IR systems aim to provide accurate and meaningful results by matching the user's information needs with the most relevant documents.

Here are the key components and techniques involved in information retrieval in NLP:

1. Indexing: In IR, documents are first processed and indexed to create a structured representation that allows for efficient retrieval. The indexing process involves tokenizing the documents into words or terms, removing stop words (common words with little semantic value), and creating an inverted index that maps terms to the documents in which they occur.

2. Query Processing: When a user submits a query, it goes through a similar preprocessing step as the documents. The query is tokenized, stop words are removed, and the resulting terms are used to search the index for relevant documents. Query expansion techniques can also be applied to improve the recall and precision of the retrieved results.

Information Retrieval (Extraction)

3. Retrieval Models: Various retrieval models are used to rank and score the documents based on their relevance to the query. Common models include the Vector Space Model (VSM), Okapi BM25, and language models. These models consider factors like term frequency, document length, and term weighting to determine the relevance scores.

4. Relevance Ranking: Once the retrieval models calculate relevance scores for the documents, they are ranked in descending order of relevance. The top-ranked documents are considered the most relevant to the user's query and are typically presented as search results.

5. Evaluation Metrics: To assess the performance of an IR system, evaluation metrics are used, such as precision, recall, and F1 score. Precision measures the proportion of retrieved documents that are relevant, recall measures the proportion of relevant documents that are retrieved, and the F1 score combines precision and recall into a single metric.

Information Retrieval (Extraction)

6. Query Expansion: Query expansion techniques aim to enhance the user's query by adding additional terms or concepts related to the original query terms. This helps to capture a wider range of relevant documents and improve the search results.
7. Snippet Generation: In search engines, snippets or short summaries are often generated to provide a preview of the content of the retrieved documents. Snippet generation techniques extract relevant sentences or phrases from the documents that best represent the information related to the query.
8. Relevance Feedback: Relevance feedback allows users to provide feedback on the retrieved results, indicating which documents are relevant or not. This feedback can be used to refine and adjust the retrieval process, improving the relevance and accuracy of future searches.

By leveraging techniques such as indexing, query processing, retrieval models, and evaluation metrics, IR systems help users find the most relevant and valuable information to meet their information needs.

Applications of NLP

Natural Language Processing (NLP) has a wide range of applications across various fields. Here are some notable applications of NLP:

1. Sentiment Analysis: NLP is used to analyze and determine the sentiment or emotion expressed in text, such as social media posts, customer reviews, or feedback. It helps in understanding public opinion, brand reputation management, market research, and customer sentiment analysis.

2.Text Classification: NLP is employed for categorizing and organizing text into predefined categories or topics. It is used in spam detection, topic classification, document categorization, content filtering, and sentiment-based classification tasks.

3. Named Entity Recognition (NER): NLP techniques are applied to identify and extract named entities, such as names of people, organizations, locations, dates, and other specific entities, from text. NER is used in information extraction, entity linking, question answering, and knowledge graph construction.

Applications of NLP

4. Machine Translation: NLP plays a crucial role in machine translation, enabling the automatic translation of text from one language to another. It involves techniques like statistical machine translation, neural machine translation, and transformer models to improve translation accuracy and fluency.

5. Chatbots and Virtual Assistants: NLP is utilized in creating conversational agents, chatbots, and virtual assistants that can understand and respond to user queries and commands in natural language. They are employed in customer support, virtual customer assistance, voice assistants, and interactive conversational interfaces.

6. Information Retrieval: NLP techniques are used in search engines to improve the relevance and accuracy of search results. It involves analyzing user queries, understanding search intent, and retrieving relevant documents or information from large text corpora.

7.Text Summarization: NLP is applied to automatically generate concise summaries from longer texts, such as news articles, research papers, or legal documents. Text summarization aids in information extraction, content skimming, and generating key insights from large volumes of text.

Applications of NLP

8. Question Answering: NLP techniques are used to build question-answering systems that can understand and respond to natural language questions. It involves analyzing the question, retrieving relevant information, and generating concise answers.

9. Speech Recognition and Natural Language Understanding: NLP is involved in speech recognition systems that convert spoken language into written text. It also contributes to natural language understanding by enabling machines to comprehend and interpret human language, facilitating human-computer interaction and voice-controlled systems.

10. Information Extraction: NLP techniques are utilized to extract structured information from unstructured text data. It involves tasks like extracting relationships between entities, identifying key facts, and populating knowledge bases with structured data.

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